

Math for Machine Learning

Week 1 – Mathematics for ML Practice Set

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Objective: Build practical understanding of mathematical ideas repeatedly used in Machine Learning. The focus is on meaning, calculation, and connection to ML rather than memorizing formulas.

1. Mean, Median and Descriptive Statistics

Mean is the average of all values.

Formula: Mean = $(x_1 + x_2 + \dots + x_n) / n$

Median is the middle value after arranging observations in ascending order. If there is an even number of observations, the median is the average of the two middle values.

ML connection: Mean and median help summarize numerical features.

2. Variance

Variance measures how far values are spread around the mean.

Population formula: Variance = $\sum (x_i - \mu)^2 / N$

A larger variance indicates greater spread, while a smaller variance indicates values are closer to the mean.

ML connection: Variance helps us understand feature variability.

3. Standard Deviation

Standard deviation is the square root of variance.

Formula: $\sigma = \sqrt{\text{Variance}}$

It represents the typical spread of observations around the mean.

ML connection: It is important for understanding feature scale and standardization.

4. Covariance

Covariance indicates how two variables change together. Positive covariance means they tend to increase together; negative covariance means one tends to increase when the other decreases.

ML connection: Covariance helps identify relationships between features.

5. Correlation

Correlation measures the strength and direction of a relationship between two variables. Its value lies between -1 and +1.

+1 → strong positive relationship

0 → no linear relationship

-1 → strong negative relationship

ML connection: Correlation can help identify relationships between features and a target. Correlation does not by itself prove causation.

6. Probability

Probability represents how likely an event is to occur.

$P(A) = \text{Number of favorable outcomes} / \text{Total number of possible outcomes}$

Conditional probability: $P(A|B) = P(A \cap B) / P(B)$

ML connection: Probability is used in classification, uncertainty estimation, Bayesian methods, and probabilistic models.

7. Vectors

A vector is an ordered collection of numerical values.

Example: $v = [2, 4, 6]$

Vectors can represent data points or feature values.

ML connection: A person's features can be represented as a vector and supplied to a machine-learning model.

8. Matrices

A matrix is a rectangular arrangement of numbers organized into rows and columns.

Example:

$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

Matrix dimensions are written as rows $\times$ columns.

ML connection: Datasets can be represented as matrices, where rows are observations and columns are features.

9. Dot Product

For vectors a and b :

$$a \cdot b = a_1b_1 + a_2b_2 + \dots + a_nb_n$$

Example:

$$[2, 3] \cdot [4, 5] = (2 \times 4) + (3 \times 5) = 23$$

ML connection: Dot products are used in weighted sums in linear models and neural-network layers.

10. Derivatives

A derivative measures how a function changes when its input changes.

Example: If $f(x) = x^2$, then $f'(x) = 2x$.

ML connection: Derivatives help determine how model loss changes when parameters change.

11. Gradients

A gradient is a collection of derivatives for a function with multiple parameters. It indicates the direction in which a function increases most rapidly.

ML connection: The gradient of loss with respect to model parameters helps determine parameter updates during training.

12. Gradient Descent

Gradient descent is an optimization method used to reduce model loss.

Update rule: $\theta_{\text{new}} = \theta_{\text{old}} - \alpha \nabla J(\theta)$

Where θ = model parameters, α = learning rate, and $\nabla J(\theta)$ = gradient of the loss.

The process repeatedly calculates loss, finds the gradient, and updates parameters toward lower loss.

ML connection: Gradient descent is a core optimization idea used to train many machine-learning and deep-learning models.

Quick ML Connection Summary

Final Reflection

Topic	ML Connection
Mean / Median	Dataset summary
Variance / Standard Deviation	Feature spread and scaling
Covariance / Correlation	Relationships between variables
Probability	Classification and uncertainty
Vectors	Represent features/data points
Matrices	Represent datasets and transformations
Dot Product	Weighted sums in ML models
Derivatives	Measure change in loss
Gradients	Direction for parameter updates
Gradient Descent	Optimize parameters by reducing loss